

Table 1. Main Results on Utility Metrics (Llama-3.2-3B-Instruct)

Utility is accuracy on 10 closed-ended benchmarks ($\uparrow$). Avg is the mean across the 10 tasks. RP (Retained Performance) is utility preserved relative to the Non-private / plaintext baseline. $\dagger$: continuation-style scoring. fastProve rows: $n=200$ per task, BF16, RTX 5090; ASR not evaluated.

Model	Method	Conf.	Priv.	Utility Metrics (Accuracy $\uparrow$)										Avg	RP
			ASR $\downarrow$	ARC-E	PIQA	MNLI	SST2	ANLI1	ANLI2	ANLI3	WiC	HellaS †	MMLU †		
<i>OSNIP paper (Cao et al.), full evaluation protocol</i>															
Llama-3.2-3B-Instruct	Non-private	/	1.000	0.712	0.768	0.542	0.867	0.440	0.401	0.425	0.497	0.716	0.606	0.597	100%
	Cape	$\epsilon=14$	0.404	0.490	0.613	0.461	0.642	0.366	0.345	0.365	0.486	0.324	0.245	0.434	72.70%
	DynText	$\epsilon=2.5$	0.807	0.266	0.517	0.321	0.511	0.349	0.339	0.348	0.480	0.275	0.243	0.365	61.14%
	InferDPT	$\epsilon=6$	0.667	0.269	0.509	0.331	0.508	0.352	0.320	0.333	0.491	0.271	0.239	0.362	60.65%
	OSNIP (Ours)	/	0.021	0.707	0.766	0.535	0.825	0.431	0.407	0.410	0.513	0.709	0.572	0.588	98.49%
<i>fastProve (this work), same 10 tasks, n=200, likelihood MC; RP vs our plaintext</i>															
Llama-3.2-3B-Instruct	Plaintext	/	—	0.535	0.460	0.480	0.970	0.475	0.395	0.350	0.505	0.465	0.295	0.493	100%
	Structural	/	—	0.530	0.460	0.485	0.965	0.480	0.400	0.350	0.505	0.465	0.295	0.494	100.1%
	Full noise	/	—	0.505	0.460	0.495	0.950	0.415	0.355	0.350	0.505	0.455	0.315	0.481	97.46%

WikiText-2 PPL	Non-private / Plain	Obfuscated	Relative Δ
OSNIP paper	9.63	12.58	+31%
fastProve structural	11.59	11.61	+0.12%
fastProve full	11.59	11.73	+1.15%

Notes. Absolute accuracies of fastProve plaintext may differ from OSNIP Non-private due to prompt template, scoring harness, and sample size ($n=200$). RP compares each method to its own plaintext baseline under the same protocol. Privacy ASR is reported only for OSNIP paper methods; fastProve privacy attack is out of this table’s scope. Source: OSNIP Tables 1–2; `results/raw/osnip_style_bench_n200.json`.